

Chia-Heng Liu

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EDUCATION

Cornell University

Ithaca, NY

Master of Engineering in Electrical and Computer Engineering

Exp. Dec. 2026

Topic: Mixed-Precision Hardware Accelerator for Edge LLM Inference

Advisor: Prof. Mohammad Alian

Coursework: Complex Digital ASIC Design (Tapeout Course), Advanced Computer Architecture (Multicore),

Memory-Centric Computing, Computer Architecture, Data Center Architecture, Computer Networks

National Tsing Hua University (NTHU)

Hsinchu, Taiwan

Bachelor of Science in Materials Science and Engineering

Sep. 2020 – Jun. 2024

Coursework: Logic Design, Computer Organization, Electronics, Signals and Systems, Semiconductor Design

SELECTED PROJECTS

ASIC Project: Mixed-Precision Hardware Accelerator for Edge LLM Inference

SystemVerilog, Synopsys (VCS, Design Compiler), Pytorch

Aug. 2025 – Present

- Designed a **transformer accelerator** with INT8 GEMM cores and a unified stochastic nonlinear unit (ReLU, GELU, SiLU, Sigmoid, Softmax, RMSNorm).
- Enabled **runtime accuracy-throughput trade-offs** via stochastic bit-stream length.
- Analyzed architectural trade-offs across precision scaling, SIMD width, throughput, and energy efficiency.

ASIC Project: Stepwise-Streaming Systolic CNN Accelerator

SystemVerilog, Synopsys (VCS, Verdi, Design Compiler), ARM (SRAM Compiler), Pytorch

May. 2025 – Jul. 2025

- Designed a systolic CNN accelerator with **fused MAC-ReLU-Max**, cached weights per PE mesh, and window buffering for prefetch; achieved **129 GOPs/s (INT8) @ 500 MHz**.
- Executed RTL-to-synthesis **ASIC flow (TSMC 40 nm)** using DC/VCS/Verdi, achieving **412.7K μm** , validated with cycle-accurate testbenches and a **quantized MNIST demo**.

CPU Project: 4-Core Pipelined RISC-V CPU with 2-Way Write-Back Cache

SystemVerilog, Verilator, PyMTL3

Sep. 2025 – Nov. 2025

- Designed a **5-stage, fully-bypassed 4-core RISC-V CPU** with hazard detection, squashing, and **2.5 IPC** execution; integrated a **router-based NoC** for multicore communication.
- Implemented **latency-insensitive variable-latency units** and a **2-way write-back set-associative cache (LRU)**; verified correctness and performance using **PyMTL3** under multithreaded workloads.

EXPERIENCE

Research Assistant at NYCU Cerebral and Reliable SoC Laboratory

Jul. 2024 – Aug. 2025

First-Author Publication: Energy-Efficient and Accurate Stochastic Computing-based

Multiply-Accumulate Architecture for Neural Network Accelerator Design

IEEE MCSoc 2025

- Developed a stochastic-computing MAC for energy-efficient DNN inference, synthesized in **TSMC 40 nm** (1049 μm^2 , 0.645 mW), achieving **90.6% lower MSE** and **96.8% MNIST accuracy**.

Co-op Project with ITRI: NoC Accelerator Simulator Design and Low-Power Data Transmission Protocol, Movement, and Reuse Method

Jul. 2024 – Jan. 2025

- Built a NoC-based AI accelerator simulator by integrating **Gem5 (RISC-V)** and **Noxim**, enabling multicore scheduling and dependency-aware data reuse analysis.

TECHNICAL SKILLS

Hardware: Verilog, SystemVerilog, SystemC, RISC-V, Network-on-Chip

EDA Tools: Synopsys (VCS, Verdi, Design Compiler, DesignWare), ARM (SRAM Compiler), Cadence (Innovus), Intel ModelSim, Xilinx Vivado, Verilator

Programming: C++, Python, Pytorch, RISC-V Assembly, Linux, Shell, TCL, LaTeX